



Approximation Algorithms for Big Data

Lecture 12: Fast JL Transform & Class Conclusion

Spring 2026

Instructor: Tal Wagner

Administrative Announcements

- HW3: due **tonight**
- Class engagement reporting: due by **July 17**
- Proposal re-submission (if original proposal was declined or not submitted by now): due **July 21**

Administrative Announcements

- **Last call** - University-wide teaching survey:
- https://my.tau.ac.il/TAU_Student/TeachingSurvey
- Your feedback is highly valuable
- **Still not at minimum number of responders**

הדקנט לחדשנות
בהוראה ובלמידה
אוניברסיטת תל אביב



סקר שביעות רצון מהוראה
סמסטר ב', תשפ"ז

יפתח מחר, 23/6 בשעה 8:00 למענה

ספרו לנו, איך עבר הסמסטר?



מקנה
נקודות
בבידינג

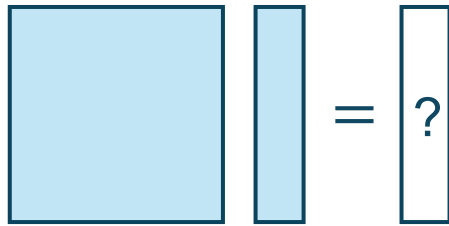
הקליקו לסקר

כניסה למידע האישי > מענה על הסקר

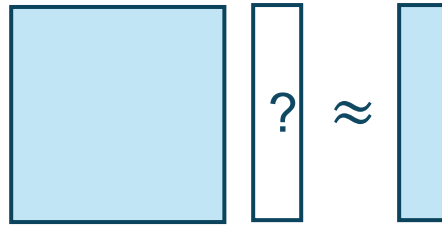
הסקר יימשך שלושה וחצי שבועות
ויהיה פתוח למענה עד ל-18/7/26

Recap: Sketch-Based Linear Algebra

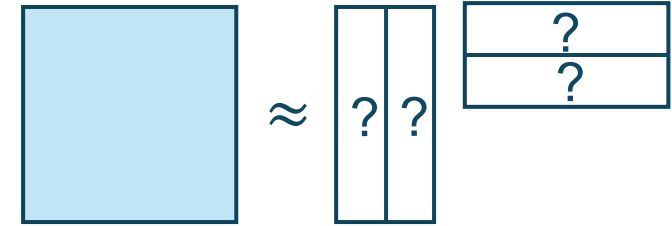
- “Sketch-and-solve” paradigm:
 - Approximate matmul
 - Approximate regression
 - Approximate LRA
- JL and JL-like transforms lead to sketching-based linear algebra algorithms



**Matrix-vector /
matrix-matrix
multiplication**

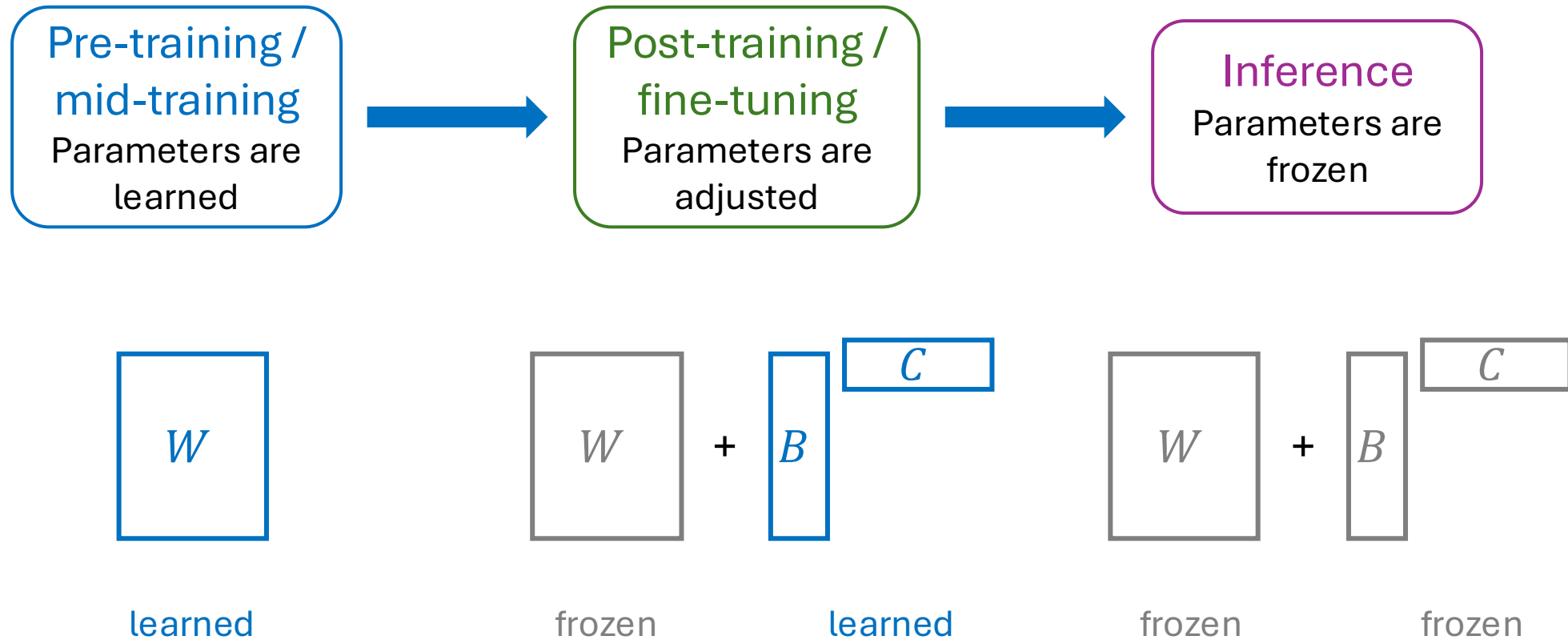


Linear regression



**Low-rank
approximation**

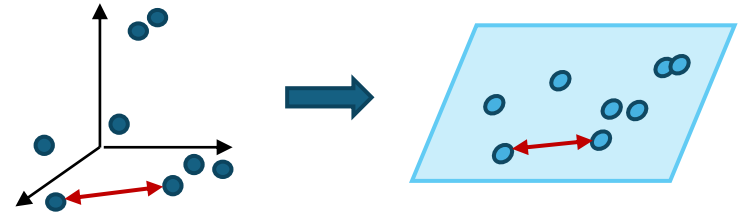
Recap: LLM Finetuning with LoRA



Fast Johnson-Lindenstrass Transform (FJLT)

Dimension reduction in nearly linear time

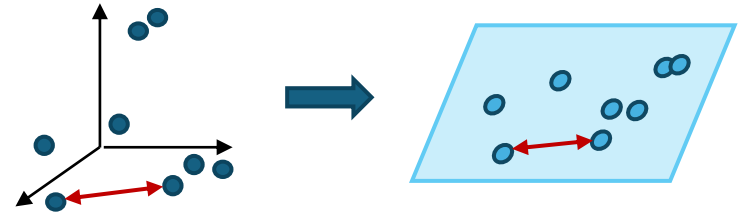
Reminder: JL and DJL



- **Lemma (DJL):** Let $\varepsilon, \delta \in (0,1)$ and $m = O\left(\frac{\log(1/\delta)}{\varepsilon^2}\right)$. Let $M \in \mathbb{R}^{m \times d}$ be a matrix with iid entries from $N(0,1)$. For every unit vector $x \in \mathbb{R}^d$,

$$\Pr \left[1 - \varepsilon \leq \left\| \frac{1}{\sqrt{m}} Mx \right\|_2 \leq 1 + \varepsilon \right] \geq 1 - \delta$$

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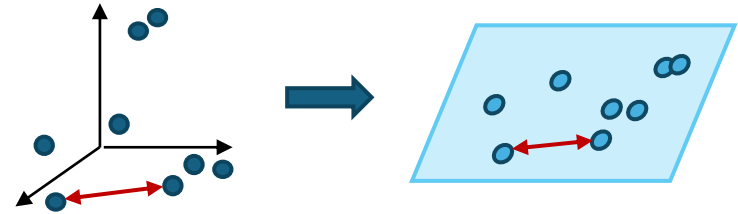


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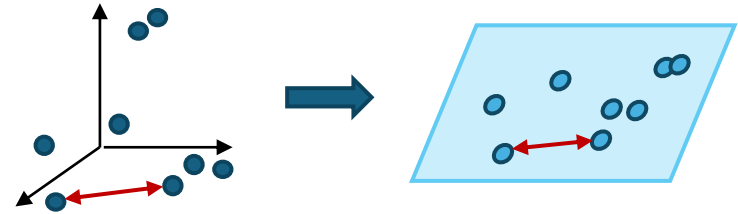


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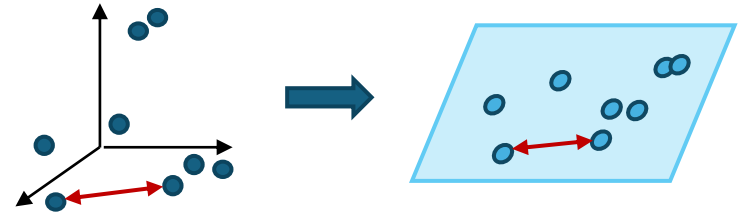


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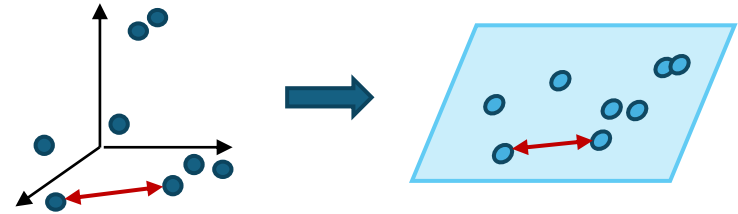


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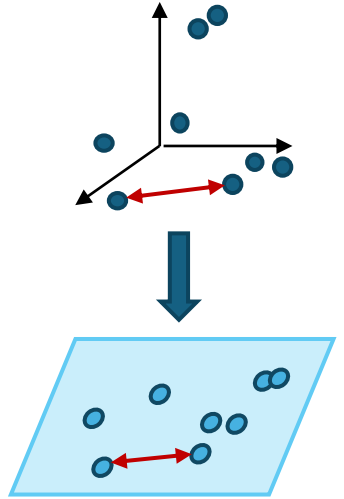


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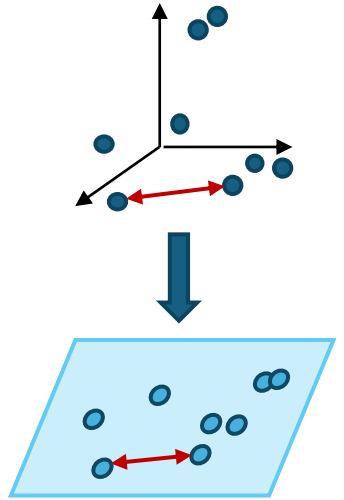
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- **Fast JL** [Ailon & Chazelle '06]: time $\tilde{O}(d + m)$ with almost the same m : $m = \tilde{O}(1/\varepsilon^2)$
- We will see:
 - Target dimension: $m = O\left(\frac{\log^2(d/\delta) \log(1/\delta)}{\varepsilon^2}\right)$
 - Time per point: $O(d \log d + m)$

Intuition (1): Fast Random Projection?



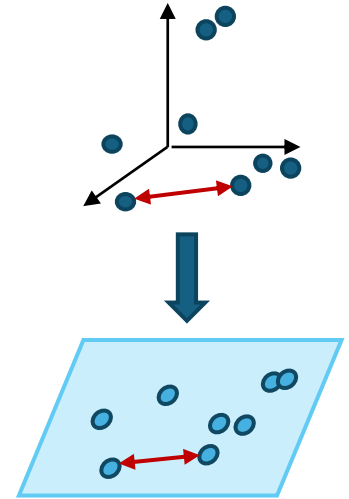
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- Goal: Compute Mx given $M \in \mathbb{R}^{m \times d}$ and $x \in \mathbb{R}^d$
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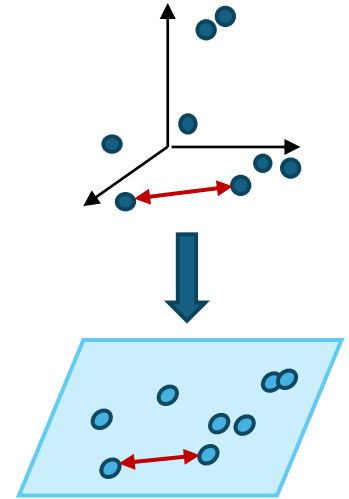
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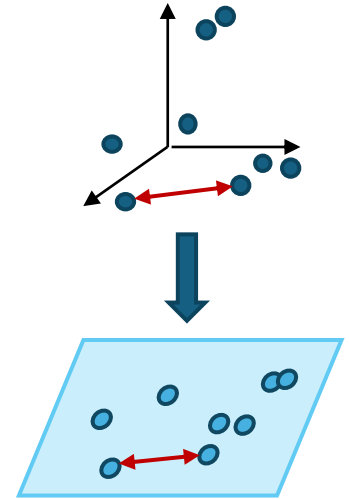
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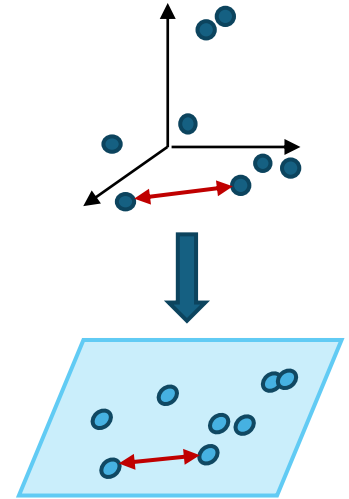
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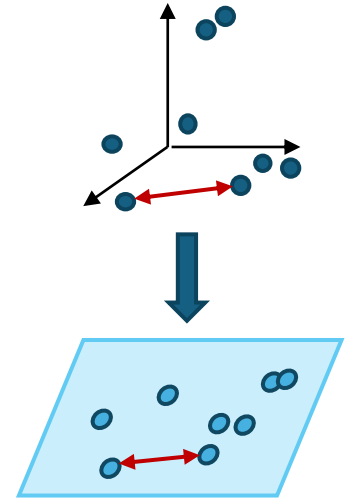
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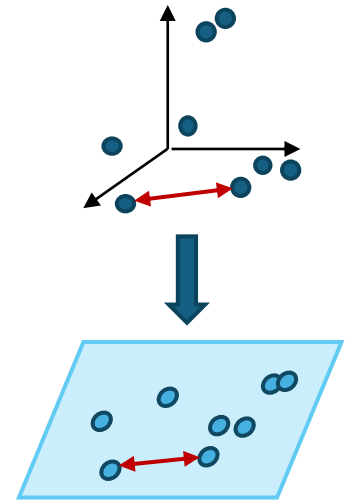


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 - Sparsity: e.g., what if M is diagonal?
 - Non-sparse structure: e.g., what if all rows of M are the same?
- But we also need M to “simulate a random projection”

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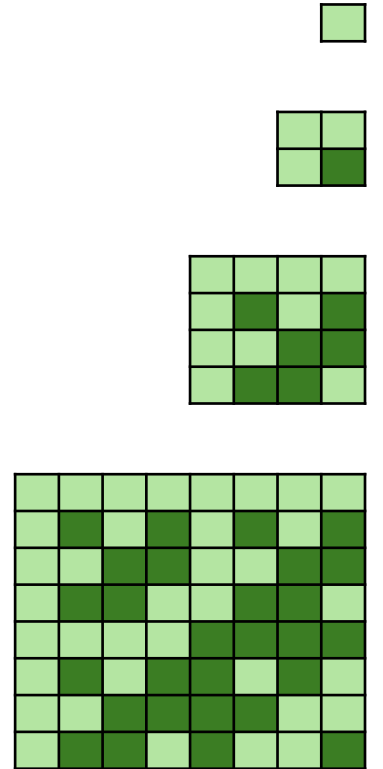
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- **Upshot:** if we could flatten the coordinates of x , to as close to $\|x\|_2^2 / \sqrt{d}$ as possible, we could uniformly sample few coordinates for dimension reduction

The Hadamard Matrix

- The Hadamard matrix of order $2^\ell \times 2^\ell$ is defined by induction:

$$H_{2^0} = [1], \quad H_{2^\ell} = \frac{1}{\sqrt{2}} \begin{bmatrix} H_{2^{\ell-1}} & H_{2^{\ell-1}} \\ H_{2^{\ell-1}} & -H_{2^{\ell-1}} \end{bmatrix}$$

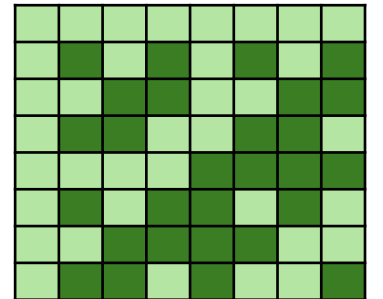
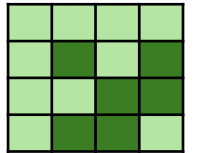


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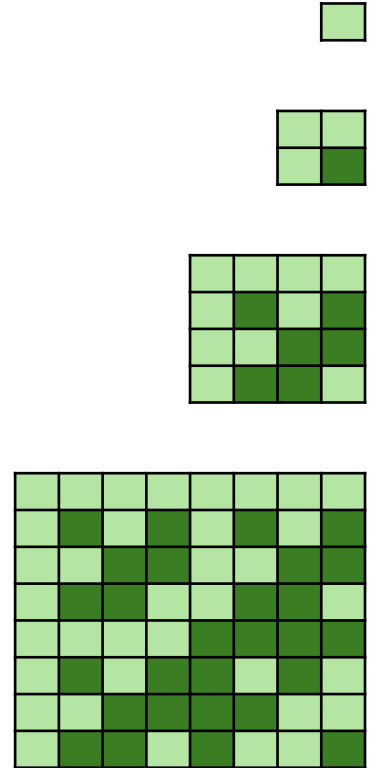


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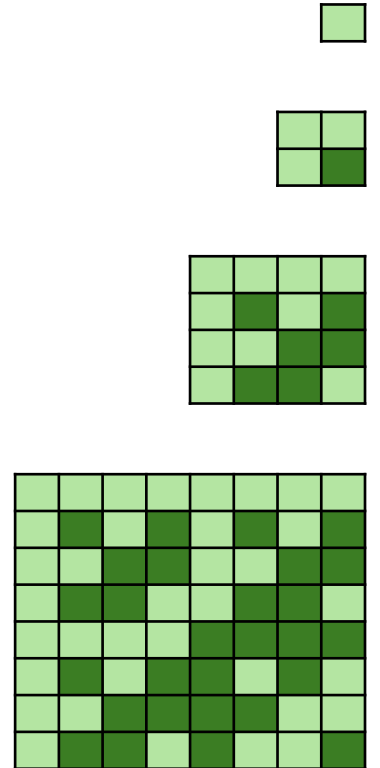


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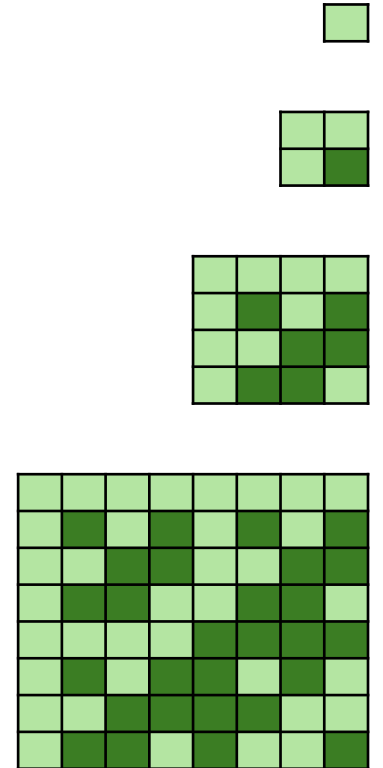
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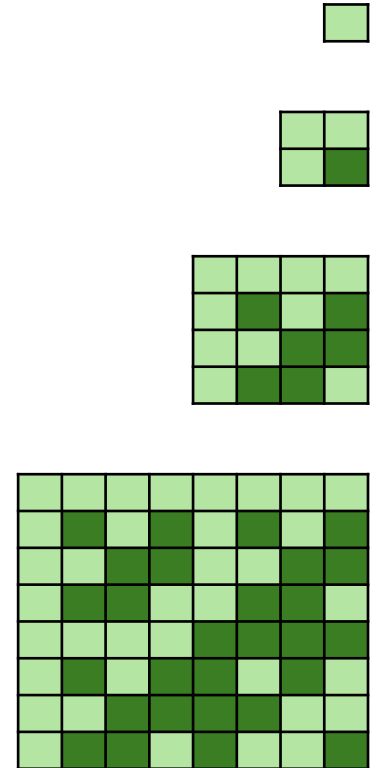


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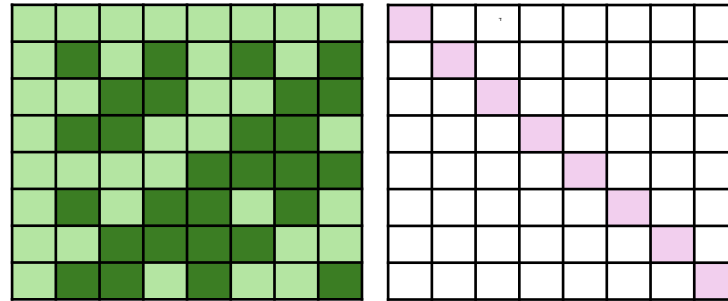
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 - Hence: $\|Hx\|_2 = \|x\|_2$ for all $x \in \mathbb{R}^d$ (**proof?**)
 - Given $x \in \mathbb{R}^d$, we can compute Hx in time $O(d \log d)$ (**how?**)



Randomized Hadamard Transform (RHT)

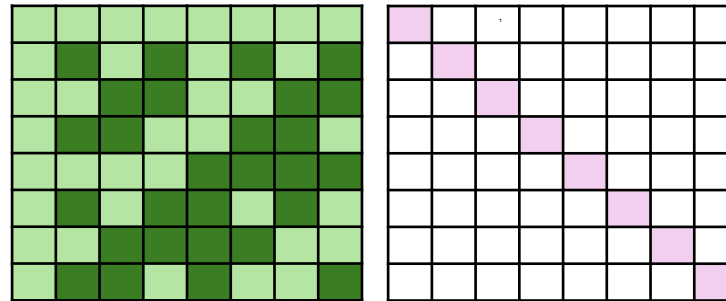
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- **RHT:** H times a diagonal matrix D with iid random signs



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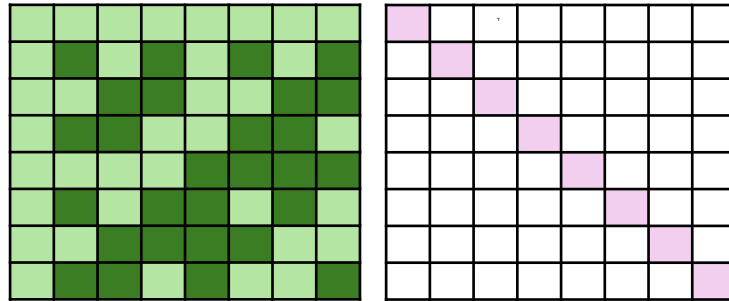
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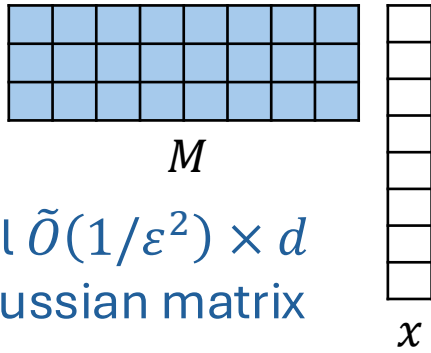
- **RHT:** H times a diagonal matrix D with iid random signs



- **Observe:** Given $x \in \mathbb{R}^d$, computing HDx takes time $O(d \log d)$
- **Lemma (flattening):** HDx has small coordinates
 - We will formalize and prove

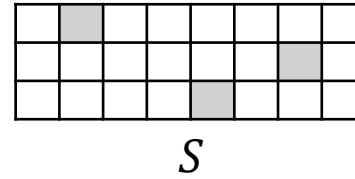
Fast JL via RHT

JL

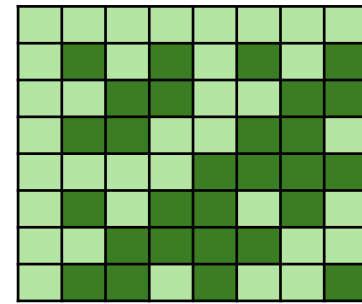


Full $\tilde{O}(1/\varepsilon^2) \times d$
Gaussian matrix

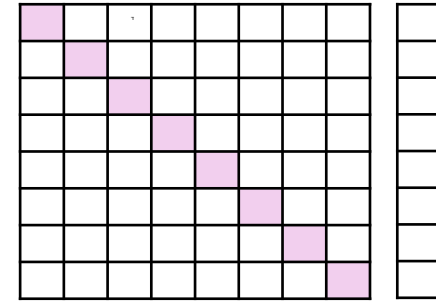
FJLT



$\tilde{O}(1/\varepsilon^2) \times d$
row-sampling
matrix



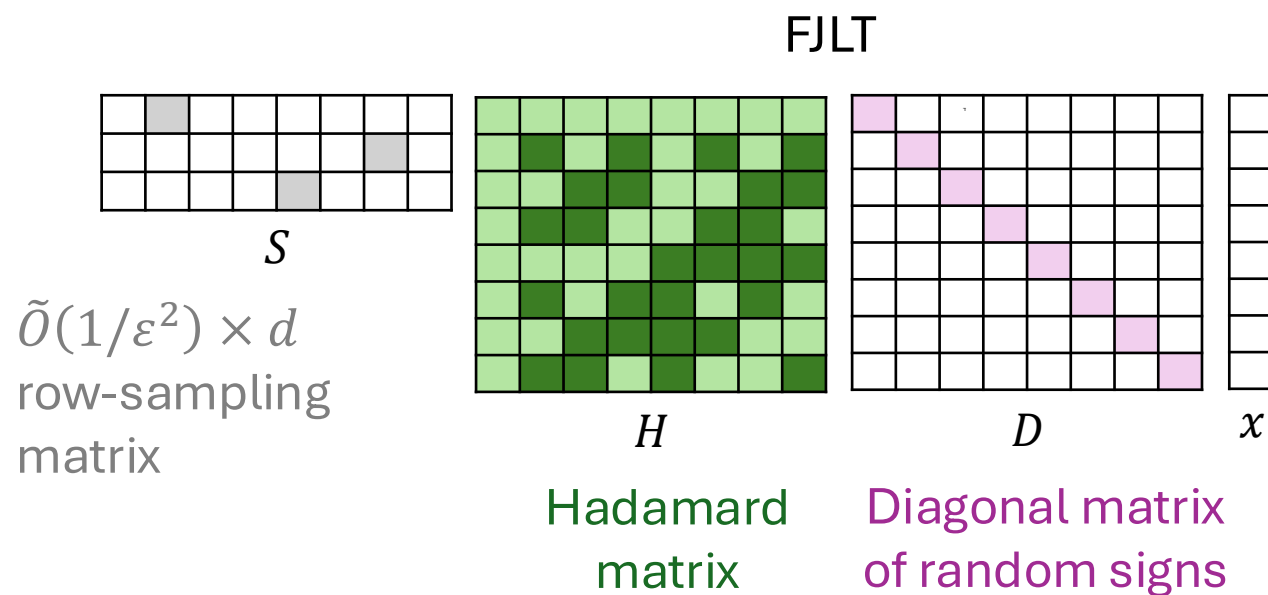
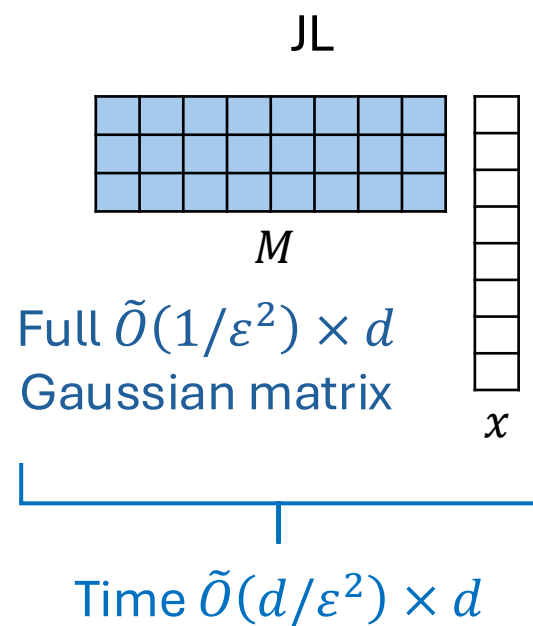
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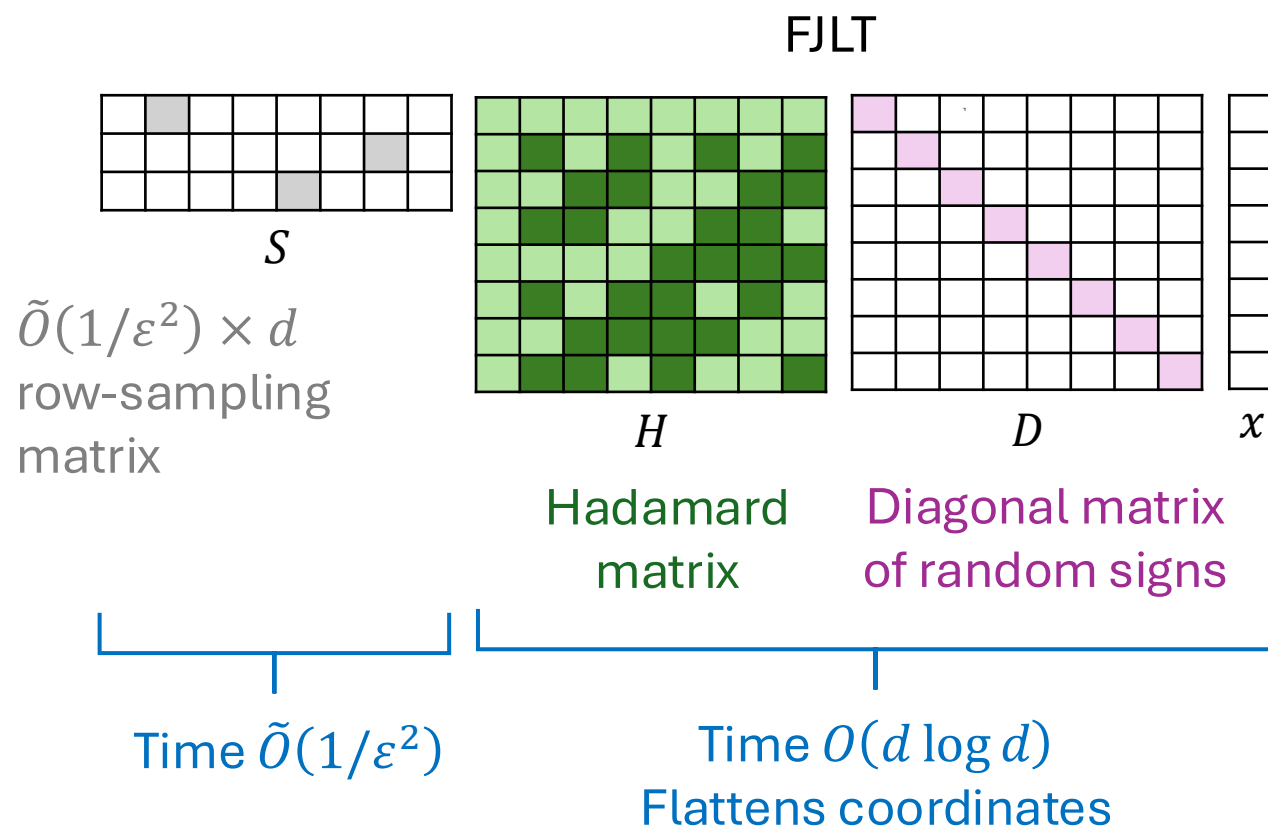
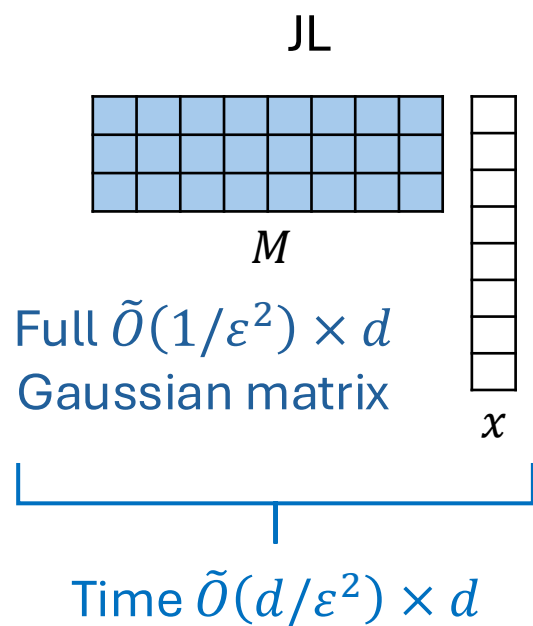
Diagonal matrix
of random signs

x

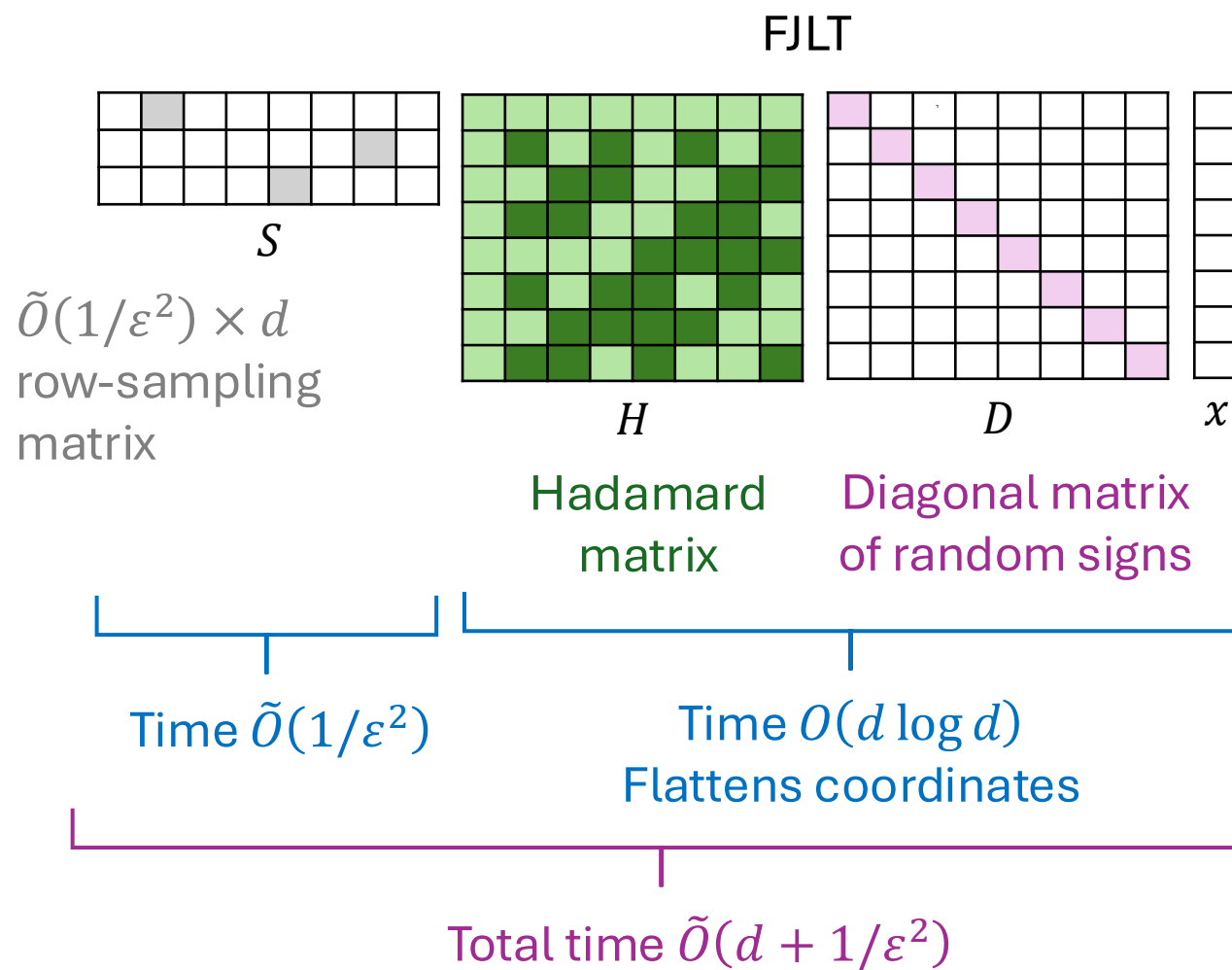
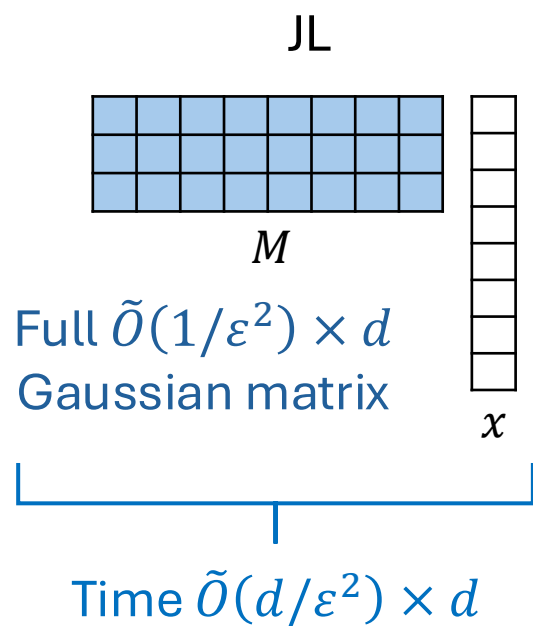
Fast JL via RHT



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RHT Flattening Lemma (1/2)

- **Lemma:** Let $x \in \mathbb{R}^d$ be a unit vector. Let $y = HDx$. Then

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- Proof:
 - Fix $i \in [d]$

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**What
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- The lemma follows by a union bound over all $i \in [d]$. 

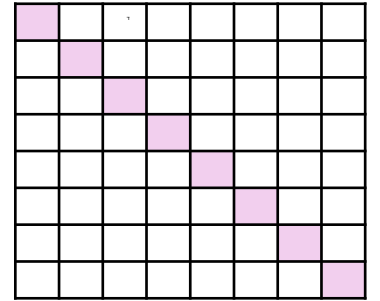
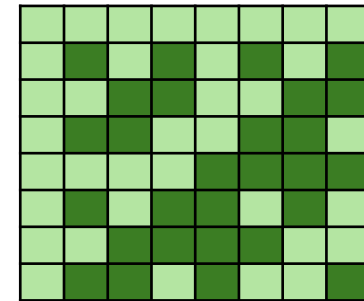
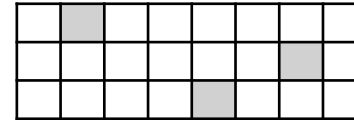
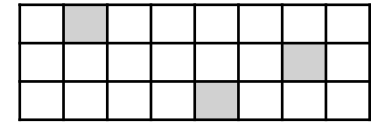
(Fast) DJL via Flattening

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 - Each row has $\sqrt{\frac{d}{m}}$ at an iid uniformly random column
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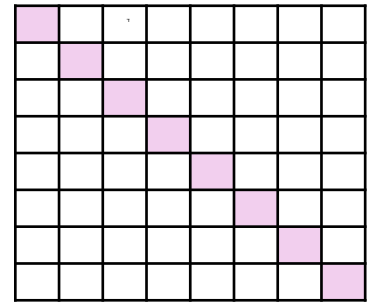
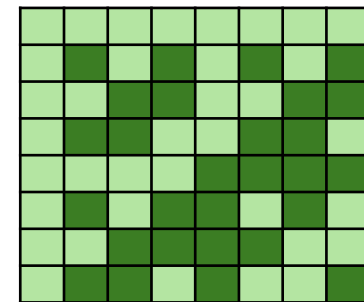
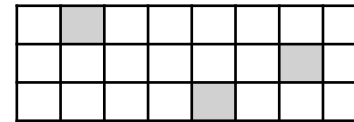
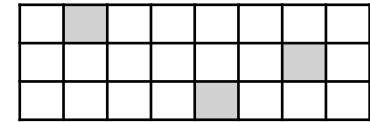
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- Lemma (DJL with SRHT):** Let $\varepsilon, \delta \in (0,1)$ and $m = O\left(\frac{\log^2(d/\delta) \log(1/\delta)}{\varepsilon^2}\right)$. Let $M = SHD \in \mathbb{R}^{m \times d}$ be an SHRT. For every unit vector $x \in \mathbb{R}^d$,

$$\Pr[1 - \varepsilon \leq \|Mx\|_2 \leq 1 + \varepsilon] \geq 1 - \delta$$

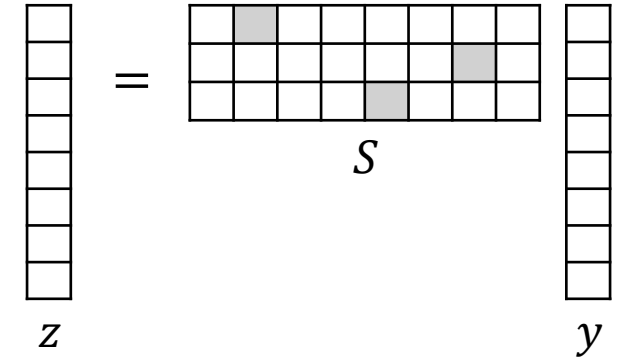
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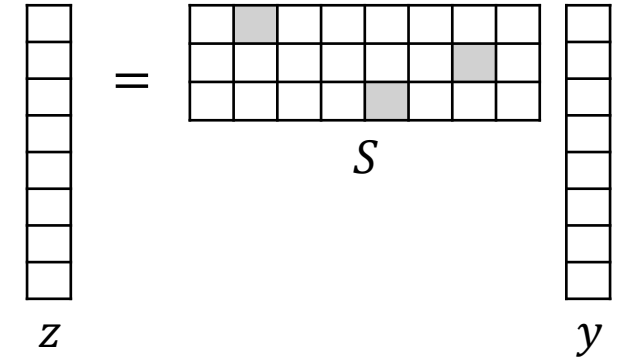
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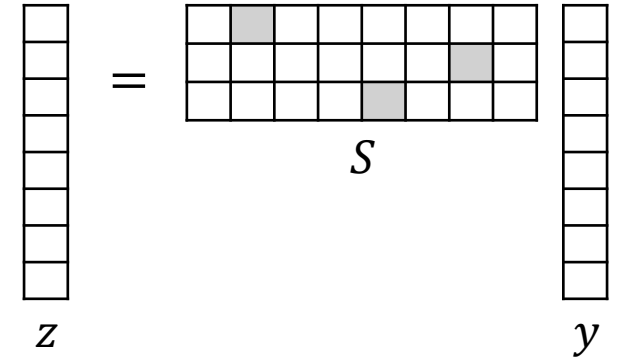
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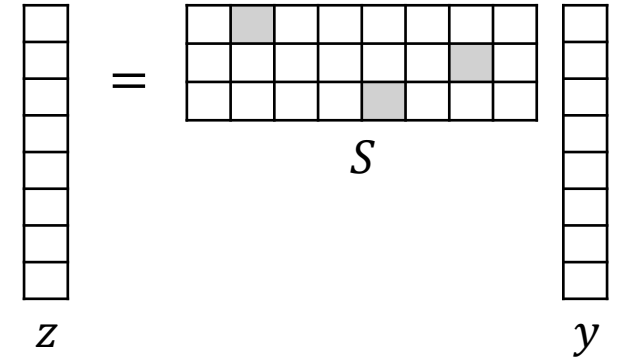
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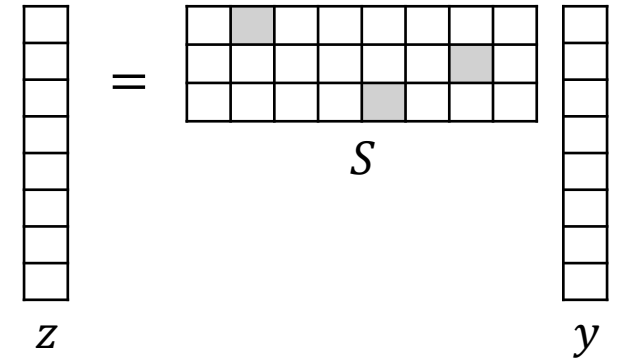
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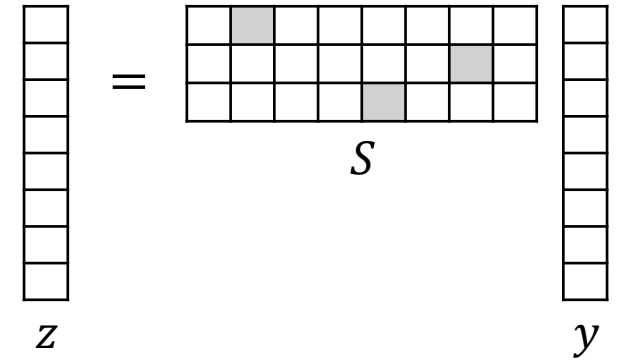
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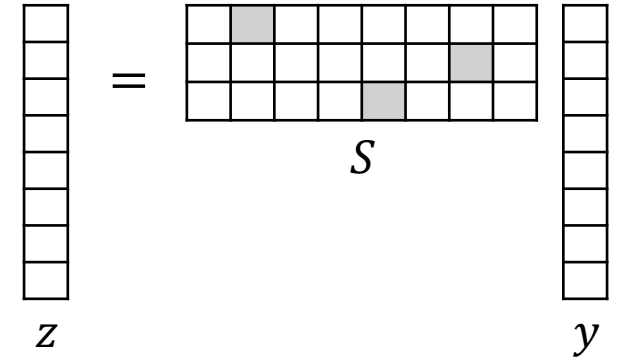
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- Hence, $m = O\left(\frac{\log^2(d/\delta) \log(1/\delta)}{\varepsilon^2}\right)$ suffices. 

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 - Variants of the construction with the same overall idea
 - More careful probabilistic analysis
 - Still active line of research
- RHTs and SRHT are useful in many other contexts
 - Random features for kernels
 - LSH
 - Other cases where we need a fast proxy for a random rotation/projection

Questions?

Next: Class Conclusion

(From Lecture 1)

What this class is about:

- **The foundations of randomized approximation algorithms that scale to large and high-dimensional datasets**
 - **Theory:** *(the focus)*
 - Learn algorithmic methods
 - Formulate them mathematically and prove they work
 - **Practice:** *(we'll touch on it)*
 - Explore how they are implemented and how they behave on real data
 - Explore where theory and practice connect, and when they don't
- **Conceptual goal:** Distill the **algorithmic idea** behind the math & code

What We Covered

- Metric embeddings and dimension reduction: JL, Fast JL
- Streaming algorithms: AMS tug-of-war, derandomization
- Sketching similarities: MinHash for Jaccard similarity, SimHash for angular similarity
- Nearest neighbor search
 - Theory: Locality sensitive hashing (LSH)
 - Practice: IVF (k-means), graph-based methods (HNSW)
- Probabilistic embeddings into trees: high-dimensional quadtrees, efficient EMD
- Efficient kernel methods: Random Fourier features, LSH features, hashing-based estimators
- Efficient linear algebra: matrix multiplication, regression, low-rank approximation via sketch-and-solve
- Applications to LLMs: deduplication, long-context Attention, KV-cache compression

Key Takeaways: Algorithmic Concepts

Randomized sketching of metric properties

Approximate
embeddings

Dimension
reduction

LSH

Linear
sketching

Random
features

Key Takeaways: Beyond Sketching

Concepts and techniques you are likely to meet in many other algorithmic contexts:

- Metric spaces
- Probabilistic analysis
 - Unbiased estimators, boosting by repetitions, concentration inequalities, MoM, importance sampling, ...
- Linear algebra
 - Matrix norms, orthogonal projections, pseudo-inverses, SVD, ...

What's Next?

- Current research in this area:
 - **Foundations:** almost none of the problems we saw are fully solved
 - **Theory vs. practice:** can we narrow the gap?
 - **LLMs and AI:** can this field be truly practically advantageous for large-scale AI?

Thank You

For participating in the class and **helping to create it**